Mohsen Ghafoorian

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Profile

Senior Staff research scientist, team lead and manager with 15+ years in machine learning and computer vision, currently focused on efficient on-device generative and video diffusion models. Track record of setting research direction, building and scaling distributed teams (7–20 engineers), shipping product demos, and publishing at top venues (CVPR, ICLR, ICCV, ECCV).

Work Experience

Qualcomm, Amsterdam — 2021 – present

- 2025–now **Technical Team Lead – AI Research.** Leading 6-10 researchers/engineers on efficient on-device video diffusion models; multiple on-device demos and 5 papers in top venues.
- 2023–now **Senior Staff ML/CV R&D Engineer.**
- 2022–25 **Technical Team Lead – XR Research.** Led a distributed team of ∼20 engineers on efficient 3D reconstruction and understanding; 5 papers in top venues, 20+ patents. Designed, led, and delivered systems for on-device demos and integrated features (AndroidXR and Snapdragon Spaces) on 3D reconstruction, 3D scene understanding, light estimation, and people-and-pet breakthrough.
- 2022–now **Line Manager.** Have coached 21 ML R&D engineers and researchers.
- 2021–23 **Staff ML/CV R&D Engineer.** Efficient 3D perception algorithms for augmented/virtual reality.

TomTom, Amsterdam — 2017 – 2021

- 2017–21 **Senior ML R&D Engineer.** Deep learning expert on automated generation of HD maps for self-driving cars.
- 2019–21 **Technical Team Lead.** Led a team of engineers to expand HD map coverage.
- 2020–21 **Line Manager.** Coached a team of ML R&D engineers.
- 2020–21 **Internal Director, Atlas Lab.** TomTom-side director of the TomTom & UvA Atlas Lab with 5 Ph.D. students.

Allameh Helli 3, Tehran — 2010 – 2013

- 2010–13 **Computer Group Manager and Lecturer.** Taught C++, data structures, and algorithms and managed a team of 10 CS teachers at a school of the National Organization for Development of Exceptional Talents (NODET).

Education

- 2013–17 **Ph.D. in Machine Learning.** Radboud University, Nijmegen, The Netherlands.
- 2016–17 **Visiting Scholar.** Harvard University, Boston, United States.
- 2010–12 **M.Sc. in Artificial Intelligence.** Sharif University of Technology, Tehran, Iran.
- 2005–10 **B.Sc. in Software Engineering.** University of Tehran, Tehran, Iran.

Selected Publications

Full list on my Google Scholar page (24,000+ citations, h-index 23). Author position noted per entry.

Recent: Video Diffusion & Generative Models

1. *MobileWan: Closing the Quality Gap for Mobile Video Diffusion*, under review. (Co-first author)
2. *ReHyAt: Recurrent Hybrid Attention for Video Diffusion Transformers*, **CVPR26**. (First author)
3. *Attention Surgery: An Efficient Recipe to Linearize Your Video Diffusion Transformer*, **CVPR26**. (First author)
4. *PyramidalWan: On Making Pretrained Video Models Pyramidal for Efficient Inference*, **CVPR26**. (2nd-last author)
5. *Moalign: Motion-Centric Representation Alignment for Video Diffusion Models*, **ICLR26**. (Last author)

6. *Neodragon: Mobile Video Generation using Diffusion Transformer*, **ICLR26**. (2nd-last author)

Selected Earlier Work

1. *AnyMap: Learning a General Camera Model for Structure-from-Motion with Unknown Distortion in Dynamic Scenes*, **CVPR25**. (Last author)
2. *FastCAD: Real-Time CAD Retrieval and Alignment from Scans and Videos*, **ECCV24**. (Last author)
3. *InterroGate: Learning to Share, Specialize, and Prune Repr. for Multi-task Learning*, **BMVC24**. (Last author)
4. *3D Distillation: Improving Self-Sup. Monocular Depth Est. on Reflective Surfaces*, **ICCV23**. (Last author)
5. *DG-Recon: Depth-Guided Neural 3D Scene Reconstruction*, **ICCV23**. (Last author)
6. *EL-GAN: Embedding Loss Driven Generative Adversarial Nets for Lane Detection*, **ECCVW18**. (First author)
7. *Transfer Learning for Domain Adaptation in MRI: Application in Brain Lesion Segmentation*, **MICCAI17**. (Co-first author)
8. *Location-Sensitive Deep Convolutional Neural Networks for Segmentation of White Matter Hyperintensities*, **Nature Sci. Reports 2017**. (First author)
9. *A Survey on Deep Learning in Medical Image Analysis*, **MedIA 2017**. (Co-author, 20k+ citations)
10. *Automatic Abstraction in Reinforcement Learning using Ant System Algorithm*, **AAAI13**. (First author)

Academic Supervision

- **Aritra Bhowmik**, Ph.D. Intern, U. Amsterdam: physical plausibility of video diffusion models. **ICLR26**.
- **Andrea Porfiri**, Ph.D. Intern, Politecnico di Milano: general camera model for dynamic SfM. **CVPR25**.
- **Florian Langer**, Ph.D. Intern, U. Cambridge: contrastive retrieval for CAD-based 3D reconstr. **ECCV24**.
- **Xuepeng Shi**, Ph.D. Intern, Imperial College London: reflective surfaces in self-supervised depth. **ICCV23**.
- **Osman Ulger**, Ph.D. student, U. Amsterdam: structured semantic segmentation. **BMVC22**.
- And 6+ additional M.Sc./B.Sc. theses and research internships (including an ICDIP 2021 Best Paper and a SPIE 2016 best-thesis award).

Honors and Awards

2021	**Best Paper Award**, International Conference on Digital Image Processing (ICDIP).
2016	**Research-visit grant** (9K USD) from the Surgical Planning Laboratory, Harvard, and the annual travel grant of the Dutch MS Research Foundation.
2010	**27th** rank (top 0.3%) in the A.I. Master's program entrance exam, among ∼10k participants.
2005	**448th** rank (top 0.2%) in the Bachelor's program entrance exam, among ∼300k participants.

Service, Teaching & Skills

Open Source	Public model releases on Hugging Face; **MobileWan** demo featured by Hugging Face's open-source team on ZeroGPU infrastructure.
Peer Review	Reviewer for ICLR, NeurIPS, BMVC, IJCNN.
Teaching	Lecturer/TA for 10+ B.Sc./M.Sc. courses at Radboud, Sharif, and U. Tehran – including *Intelligent Systems in Medical Imaging*, *Machine Learning in Practice*, *Computer-Aided Diagnosis*, *Artificial Intelligence*, and *Data Structures & Algorithms* (medical-imaging M.Sc. course rated 9/10 median). Invited lecturer, *Deep Learning Under the Hood*, NA-MIC Project Week, CSAIL MIT (rated 4.3/5).
Leadership	Research direction, team building and scaling, cross-site coordination, technical mentoring and people management.
Technical	PyTorch, TensorFlow, Python/C++; distributed training, model efficiency (quantization, pruning, distillation), on-device/edge deployment.
Basics	Dutch-Iranian. Languages: Persian (native), English (fluent), Dutch (B1).